GUIDED NOTES: GRAPHS OF EXPONENTIAL FUNCTIONS (EL2)

Section 3.2.1

Name: _____ Date: _____ Class: _____

Learning Objective: Graph exponential functions and determine the domain, range, and asymptotes.

Key Concepts / Cues	Lecture Notes & Examples
Activity 1	Consider the function $f(x) = 2^x$. **(a)** Fill in the table of values for $f(x)$. Then plot the points on a graph.

x	$f(x)$
-2	
-1	
0	
1	
2	

(b) What seems to be happening with the graph as x goes toward infinity? Plug in large positive values of x to test your guess, then describe the end behavior.

(c) What seems to be happening with the graph as x goes toward negative infinity? Plug in large negative values of x to test your guess, then describe the end behavior.

(d) Complete the graph you started, connecting the points and including the end behavior you've just determined.

(e) Does your graph seem to have any asymptotes?

(f) What the equation for each asymptote of $f(x)$? Select all that apply.

(g) Find the domain and range of $f(x)$. Write your answers using interval notation.

(h) Find the interval(s) where $f(x)$ is increasing and the interval(s) where $f(x)$ is decreasing. Write your answers using interval notation.

Note

The graph of an exponential function $f(x) = b^x$ where $b > 1$ has the following characteristics:

- Its domain is $(-\infty, \infty)$ and its range is $(0, \infty)$.
- It is an exponential growth function; that is it is increasing on $(-\infty, \infty)$.
- There is a horizontal asymptote at $y = 0$. There is no vertical asymptote.
- There is a y-intercept at $(0,1)$. There is no x-intercept.

Activity 2

Consider the function $g(x) = \left(\frac{1}{2}\right)^x$.

(a) Fill in the table of values for $g(x)$. Then plot the points on a graph.

x	$g(x)$
-2	
-1	
0	
1	
2	

(b) What seems to be happening with the graph as x goes toward infinity? Plug in large positive values of x to test your guess, then describe the end behavior.

(c) What seems to be happening with the graph as x goes toward negative infinity? Plug in large negative values of x to test your guess, then describe the end behavior.

(d) Complete the graph you started in the previous example, connecting the points and including the end behavior you've just determined.

(e) What are the equations of the asymptote(s) of the graph?

(f) Find the domain and range of $f(x)$. Write your answers using interval notation.

(g) Find the interval(s) where $f(x)$ is increasing and the interval(s) where $f(x)$ is decreasing. Write your answers using interval notation.

Activity 3

Consider the two exponential functions we've just graphed: $f(x) = 2^x$ and $g(x) = \left(\frac{1}{2}\right)^x$.

(a) How are the graphs of $f(x)$ and $g(x)$ similar?

(b) How are the graphs of $f(x)$ and $g(x)$ different?

Note

We can now update so that it includes all values of the base of an exponential function.

The graph of an exponential function $f(x) = b^x$ has the following characteristics:

- Its domain is $(-\infty, \infty)$ and its range is $(0, \infty)$.
- Remember, exponential functions are only defined when $b > 0$ and $b \neq 1$ as we saw in the previous example.
  If $b > 1$, $f(x)$ is increasing on $(-\infty, \infty)$ and is an exponential growth function. If $0 < b < 1$, $f(x)$ is decreasing on $(-\infty, \infty)$ and is an exponential decay function.
- There is a horizontal asymptote at $y = 0$. There is no vertical asymptote.
- There is a y-intercept at $(0,1)$. There is no x-intercept.

Activity 4

Let's look at a third exponential function, $h(x) = 2^{-x}$.

(a) Before plotting any points or graphing, what do you think the graph might look like? What sort of characteristics might it have?

(b) Fill in the table of values for $h(x)$. Then plot the points on a graph.

x	$h(x)$
-2	
-1	
0	
1	
2	

(c) This function $h(x)$ looks to be the same as a function we looked at previously. Use properties of exponents to rewrite $h(x)$ in a different way.

(d) In addition to plotting points, we can use transformations to graph. If we consider $f(x) = 2^x$ to be the parent function, what transformation is needed to graph $h(x) = 2^{-x}$?

Note

For a reminder of transformations from 1.8. **Big idea:** Changes *outside* f act on the outputs (y), so they move the graph vertically. Changes *inside* f act on the inputs (x), so they move the graph horizontally.

	Outside: changes outputs (y), vertical	**Inside:** changes inputs (x), horizontal
Shift	$f(x)+c$: up c	$f(x+c)$: **left** c
Reflect	$-f(x)$: over the x-axis	$f(-x)$: over the y-axis
Stretch	$a\,f(x)$: vertical stretch by a	$f(ax)$: horizontal **compression** by a

Activity 5

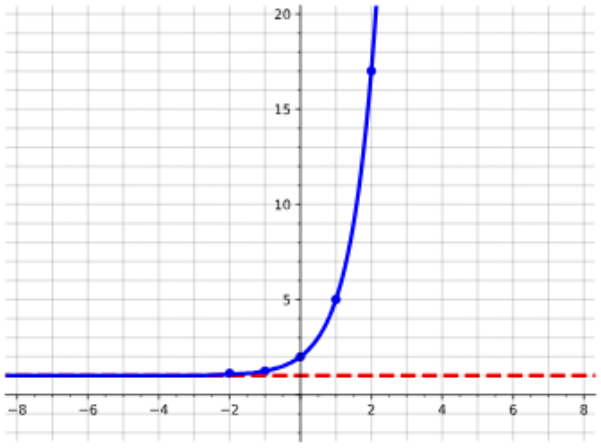
Let $f(x)=4^x$.

(a) Graph $f(x)$.

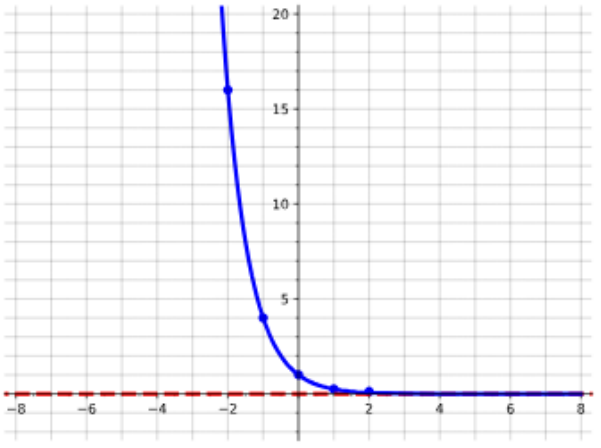
(b) Match the following functions to their graphs.

- $g(x)=-4^x$
- $h(x)=4^{-x}$
- $j(x)=4^{x+1}$
- $k(x)=4^x+1$

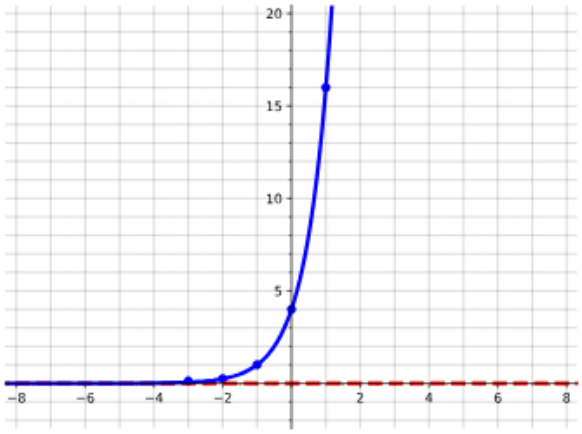
A)



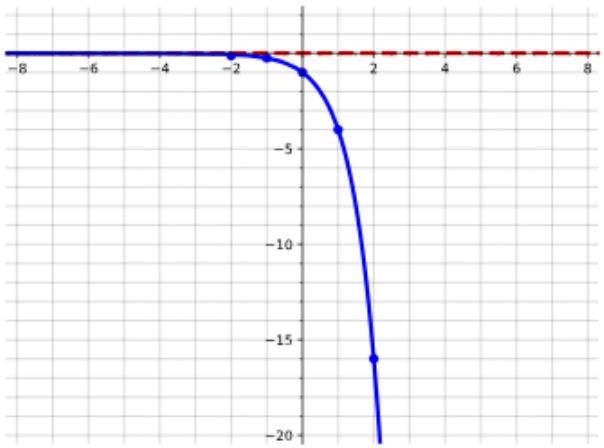
(B)



C.



D.



(c) Find the domain, range, and equation of the asymptote for the parent function $(f(x))$ and each of the four transformations $(g(x), h(x), j(x), \text{and } k(x))$.

(d) Which of the transformations affected the domain of the exponential function? Select all that apply.

(e) Which of the transformations affected the range of the exponential function? Select all that apply.

(f) Which of the transformations affected the asymptote of the exponential function? Select all that apply.

Activity 6

Consider the function $f(x) = e^x$.

(a) Graph $f(x) = e^x$. First find $f(0)$ and $f(1)$. Then use what you know about the characteristics of exponential graphs to sketch the rest. Then state the domain, range, and equation of the asymptote. (Recall that $e \approx 2.72$ to help estimate where to put your points.)

(b) Sketch the graph of $g(x) = e^{x-2}$ using transformations. State the transformation(s) used, the domain, the range, and the equation of the asymptote.

Activity 7

Graph each of the following exponential functions. Include any asymptotes with a dotted line. State the domain, the range, the equation of the asymptote, and whether it is growth or decay.

(a) $f(x) = 3^x$

(b) $f(x) = \frac{1}{3}^{x} + 4$

Practice Example 8

Solve the following problems to test your understanding of graphing exponential functions.

(a) Find the horizontal asymptote, domain, and range of the function $f(x) = 2^x - 3$. Describe the behavior of its graph.

(b) Let $f(x) = 3^x$. Write the equation of the function $g(x)$ that results from shifting $f(x)$ left by 2 units, vertically stretching by a factor of 5, and shifting up by 4 units.

(c) An exponential function has a horizontal asymptote of $y = 5$ and passes through the points $(0,8)$ and $(1,14)$. Find its equation in the form $f(x) = a \cdot b^x + k$.

SUMMARY & REFLECTIONS
(Write 2-3 sentences summarizing the key concepts of today's lecture):